Digantara AI/ML Data Annotation Assessment — Written Response

Dataset: 10 raw FITS frames, 9568 × 6380, 16-bit unsigned, single band. **Method:** scripted annotation pipeline (source in the accompanying repository). **Result:** 700 tiles of 1024 × 1024 with YOLO-Ultralytics segmentation labels, two classes — `0 = blob` (star), `1 = streak` (space object).

Annotation was scripted rather than drawn by hand because the frames contain **3 400 to 23 600 sources each**, roughly 100 000 objects in total. At that density a manual pass is not slow, it is infeasible; the engineering question is therefore how to make an automatic decision *defensible*, which is what the four answers below address.

(a) Pre-processing steps applied

Inspection first. The ten frames are **not one homogeneous set**, and the split drives every later choice:

	8 × UUID-named	**2 × `CAM_B_*`**
Background median	1–3 ADU	45–46 ADU
Maximum value	4095 (12-bit, clipped)	65535 (full 16-bit)
Header	full observatory metadata, `static_staring`	minimal, has `GAIN`, `FRAMENO`

Four measurements then determined the pipeline:

1. The faint frames are Poisson counts at λ ≈ 1. In `1a600998`, 99 % of pixels are 0, 1 or 2 ADU. The **median absolute deviation of the raw image is exactly 0** on five of the ten frames, so the textbook `median + k·MAD` threshold divides by zero. Noise cannot be estimated on raw pixels here.

2. The background is not flat. The frame centre runs **1.89× to 3.02×** brighter than the corners in all ten frames. A single global threshold would over-detect the centre and go blind at the edges.

3. There are no hot pixels. Stacking all ten frames, **zero** pixels fire above threshold in ≥ 8 of them. No bad-pixel map is needed — this is reported because it was checked, not assumed.

4. The point spread is stable, 3.5–5.2 px across frames and cameras.

The resulting steps, in order:

1. **2D background model** — block means on a 64 px mesh, median-filtered across blocks to reject blocks contaminated by bright sources, bilinearly interpolated back to full resolution, and subtracted. This answers (2).
2. **PSF-matched filter** — convolution with a Gaussian of σ = 1.1 px matching the measured point-source width. This is the load-bearing step: it is both the optimal filter for point sources *and* the thing that makes the quantised data continuous again, because averaging ~15 pixels restores a measurable noise distribution. It answers (1).
3. **Robust noise scale** — estimated from the **lower half** of the filtered histogram (`median − P15.87`). Sources only ever add flux, so they live in the upper half and cannot bias this statistic. It is deliberately treated as a *scale*, not a Gaussian σ: the filtered Poisson distribution is not Gaussian, so the operating threshold built on it is validated empirically in (d) rather than by assuming a tail probability.
4. **Display rendition** — `asinh` stretch, computed once per frame from global statistics so all 70 tiles share one transfer function and seams are invisible. `asinh` rather than linear so faint stars and saturated streaks are both visible in one image.

No flat-field or dark frame was supplied, so neither was applied; the 2D background model absorbs the large-scale component of both.

(b) Handling image boundaries when tiling to exact 1024 × 1024

9568 / 1024 = 9.34 and 6380 / 1024 = 6.23, so an exact grid does not fit. Padding invents edges the annotator must label around; dropping the remainder loses data, which the question forbids.

The grid used is non-uniform and overlapping. The last tile on each axis is pulled flush to the far edge and the rest are spread evenly between:

axis	tiles	offsets	strides	overlap
x (9568)	10	0, 949, 1899, … 8544	949 / 950	74–75 px
y (6380)	7	0, 893, 1785, … 5356	892 / 893	131–132 px

70 tiles per frame, every tile exactly 1024 × 1024, every tile fully in-bounds, zero padding, minimum coverage 1. `src/tiling.py` asserts all four properties and additionally verifies that `repatch(cut(x)) == x` **bit-exactly**, on the real frame shape plus three edge cases. The claim is then checked on the real data rather than only on synthetic arrays: `src/repatch.py --verify` reassembles each frame from its 70 exported tile files and asserts the result is pixel-identical to the frame rendered straight from the FITS. **All ten frames pass**, which is what makes the submitted full-size imagery literally repatched rather than merely re-rendered.

The more important decision is **ordering: detection runs once on the whole frame, and the finished masks are then cut into tiles.** The frame is only 61 Mpx and fits in memory, so this costs nothing — and it makes boundary handling a non-problem rather than a problem to solve. An object lying across a cut was segmented once at its true full extent; each tile simply records the part visible inside it. Detecting per tile instead would see two truncated fragments, measure both as shorter than the streak threshold, and silently relabel a streak as two blobs. Re-patching is likewise exact, because overlapping pixels come from one source frame.

(c) Distinguishing short/fat streaks from blobs

The rule. An object is a streak when, measured on its 5σ **seed core**, the major axis exceeds **3 × the frame's own PSF width** *and* the elongation exceeds **2.5**. The PSF is re-measured per frame as the median major axis of near-circular cores (3.46–5.18 px across this dataset), so both thresholds are expressed in units of that frame's point-source size rather than as pixel counts tuned on one camera.

Three things had to be fixed before that rule behaved, each found by measurement:

Geometry must be measured on the seed core, not the exported mask. The 2.5σ isophote used for mask extent traces ragged edges on faint sources, and raggedness reads as elongation. Classifying on it produced **1 649 "streaks"** in one frame whose true count is near 10. Measuring on the high-SNR 5σ core and exporting the 2.5σ extent separates "what shape is it" from "where does it end".

The object unit must be one connected core. Two stars sharing a single isophote blob give a bimodal pixel set whose second moments are long and thin — the pair reports as a streak. Making the seed core the unit of annotation, and assigning each core its isophote territory by watershed, removes this failure mode entirely.

A deblend split is believed only if its pieces are round. Bright close pairs saturate and bridge at the seed threshold, arriving as one dumbbell, and must be split. But **tumbling objects modulate in brightness along a streak**, which produces exactly the same two-peaks-with-a-saddle signature, and splitting those destroyed real streaks (longest surviving streak fell from 62.9 px to 33.6 px). The discriminator is the *result* of the split: a star pair breaks into round pieces, a modulated streak into pieces still elongated along the same axis. Any shape test applied *before* splitting is circular — it protects precisely the merged pairs it is supposed to resolve, which was confirmed by trying it.

Two hypotheses were tested and rejected, recorded here because the negative results constrain the problem:

- *Streaks in a frame share an orientation.* They do not. Across a full frame strong streaks spread over ±20–30°, and marginal candidates aligned with them 12 % of the time against 11 % expected by chance. Unusable.
- *The residual elongated bright sources are comatic* (the optics are f/1.3). They are not. Comatic flares point radially from the optical axis; the measured deviation from radial was 37–70°, indistinguishable from the control population of bright round blobs.

Known limit, reported rather than hidden. Objects in the **3–5 × PSF** band a few pixels thick cannot be separated from bright stars with asymmetric wings on single-frame morphology alone. These are labelled as streaks but carry a `marginal` flag in `summary.csv`. Resolving them properly requires cross-frame association — a source that moves between exposures is an object, one that does not is a star. That was attempted on the two `CAM_B` frames (11 minutes apart): the fields do overlap (915 of 3 821 bright sources match within 3 px, against ~39 expected by chance) but share no common translation, so the field has rotated, as an alt-az mount would. Aligning them needs an astrometric fit, and these headers carry no WCS. It is the right next step, not a step taken here.

(d) Criteria for annotating faint or small blobs as the star class

A faintness floor is a false-alarm decision, so it was **measured, not chosen**.

Sources only add positive flux. Therefore every detection made in the **negated** background-subtracted image is, by construction, a false alarm. Sweeping the threshold over both signs of the image gives the false-alarm count directly, with no assumption about the noise distribution — which matters here precisely because the noise is Poisson and not Gaussian:

threshold	real detections (+)	**false alarms (−)**
3 σ	11 317 – 90 796	**268 – 2 453**
4 σ	4 522 – 43 467	**0 – 10**
5 σ	3 404 – 23 602	**0 in all 10 frames**
6 σ	2 793 – 14 360	0 in all 10 frames
8 σ	2 043 – 9 498	0 in all 10 frames

Reproduce with `python src/calibrate.py`; it asserts the zero and exits non-zero if any frame ever shows a false alarm at the operating point.

The criterion is therefore: the faintest object annotated as a star is the faintest one at which the measured whole-frame false-alarm count is zero — a connected group of at least **5 pixels** above **5 σ** in the PSF-matched-filtered image. Both halves matter: the minimum-area requirement is what removes isolated single-pixel noise spikes, and it is the standard `DETECT_MINAREA` criterion.

There is no natural "knee" in the detection-count curve to cut at — counts fall smoothly because the field is genuinely crowded with faint stars — so a false-alarm budget is the only defensible basis for the cut, and 6σ was rejected as discarding real sources for no measured gain in purity.

Results

frame	PSF px	blobs	streaks	marginal	longest streak
1a600998	5.18	3 405	6	0	62.9
3e6beceb	4.00	9 396	7	2	53.8
4a8e6cd4	5.17	6 016	1	0	62.8
4e2cdd92	5.13	6 011	3	0	70.3
66ed4268	5.13	17 847	8	6	38.3
7030c2ac	4.21	9 720	10	3	54.9
7d7fefcb	5.18	3 500	8	0	67.9
7e5f3e1c	5.14	6 008	10	0	61.7
CAM_B…207	3.46	23 635	18	11	62.3
CAM_B…276	3.46	22 249	13	9	59.1

84 streaks and 107 787 blobs over 10 frames; 31 streaks flagged marginal. These become **130 847 polygons** across the 700 tiles: an object lying across a tile cut is recorded in each tile it touches, which is correct for per-tile labels. Every streak in every frame was inspected individually on the per-frame contact sheets in `outputs/qa/`. The marginal flag reproduces that independent visual judgement: frames whose streaks read as unambiguous report 0 marginal, while `66ed4268` — where almost every candidate looked doubtful by eye — reports 6 of 8.

Assumptions

- Sources are positive excursions above a smooth background.
- No flat-field or dark calibration frames were provided; the 2D background model absorbs their large-scale component only.
- Frames are treated independently; no cross-frame association is performed.
- Saturated cores are annotated at their measured extent, which overstates the true angular size of the brightest objects.
- The `marginal` flag is advisory. If the downstream use favours precision over recall on the streak class, those 31 objects are the ones to drop first.